the item is divisible (alternatively, if each agent has a large basket that can take many items). Imagine that each online item carries one litre of water (a divisible egg); each offline agent has a bucket (basket) of capacity one litre. The algorithm distributes an online item's water to its neighbors, where the amount of water going to each neighbor represents the fraction of the item allocated to the agent. The BALANCE algorithm (a.k.a., WATER LEVEL or WATER-FILLING [AL06]) lets the water flow to the least loaded bucket (the basket with the least amount of eggs); if there are multiple least loaded buckets, the water flows to them at an equal rate. This algorithm and its generalizations achieve the optimal $1 - 1/e$ competitive ratio in all the mentioned special cases of online submodular welfare maximization, including unweighted matching [KP00], vertex-weighted matching [BJN07], display ads [FKM+09], and AdWords [MSVV07].

In the original problems where items are indivisible, we can distribute the risk through randomized decisions. However, making independent random decisions in each round is insufficient for getting a competitive ratio better than $1/2$, the baseline set by the greedy algorithm [KVV90]. In the same paper that [KVV90] introduced the online bipartite matching problem, they also gave an elegant RANKING algorithm achieving the optimal $1 - 1/e$ competitive ratio, later generalized by [AGKM11] to vertex-weighted problem. The algorithm can be viewed as letting each offline vertex independently set a random price, and having each online vertex choose the lowest price offered by its unmatched neighbors. See §5 for a further discussion on this economic interpretation of RANKING and other online matching algorithms.

2.1 Breaking the $1/2$ Barrier in Longstanding Open Problems

Recall that for online (vertex-weighted) bipartite matching RANKING achieves an optimal competitive ratio, and in particular breaks the barrier of $1/2$. For display ads and AdWords, however, finding an online algorithm strictly better than the $1/2$-competitive greedy algorithm had remained elusive for more than a decade. Fundamentally new ideas seemed necessary because a critical invariant in the analysis of RANKING fails to hold in these two problems.

In 2020, the $1/2$ barrier was broken for both problems using a new technique called Online Correlated Selection (OCS); see §5.4 for a further discussion on this technique. [FHTZ22] introduced the concept of OCS and gave a 0.508-competitive algorithm for display ads. [HZZ20] modified the definition of OCS and applied it to the AdWords problem, and as a result, obtained a 0.501-competitive algorithm. The OCS technique has then been improved in a series of works by [SA21], [GHH+21], and [BC21]. The state-of-the-art competitive ratio for display ads is 0.536, given by a multi-way OCS algorithm by [BC21].

Despite the aforementioned progress, we remark that there is no known evidence that the $(1-1/e)$-competitive ratio cannot be achieved in display ads and AdWords. Hence, closing the gaps between the upper and lower bounds for these two problems remains an important open problem.

For the general online submodular welfare maximization problem, we recall that the simple greedy algorithm is $1/2$ competitive, and (barring surprising developments in complexity theory) no polynomial-time online algorithms can do better [KPV13]. That being said, this impossibility relied on the computational hardness of maximizing a submodular function. It would be interesting to explore online algorithms with unlimited computational capacity, because practical heuristics can often solve these optimization problems better than the worst-case approximation ratio promises, and positive results along this line may point to other special submodular functions that are computationally tractable.

	Fractional/Divisible Relaxation	Original Problem
Unweighted	$1 - 1/e$ [KP00]	$1 - 1/e$ [KVV90]
Vertex-Weighted	$1 - 1/e$ [BJN07, Section 5]	$1 - 1/e$ [AGKM11]
Display Ads	$1 - 1/e$ [FKM+09]	**0.536** [BC21]
AdWords	$1 - 1/e$ [MSVV07]	**0.501** [HZZ20]

Table 1: State-of-the-art for central online bipartite matching & allocation problems.

2.2 Stochastic Rewards and Oblivious Budget

Many online advertising platforms adopt the *pay-per-click* model. In this model, an advertiser pays each time a user clicks on its advertisement. Since these platforms cannot control the user's behavior, they resort to the next best option: modeling a user's behavior stochastically, and estimating the probability that the user clicks the advertisement, known as the *click-through-rate* (CTR). As a result of the users' stochastic behavior, the platform's revenue from assigning an impression to an advertiser is also stochastic.

[MP12] introduced online matching with stochastic rewards. They analyzed both the RANKING algorithm and a variant of the BALANCE algorithm, and showed competitive ratios better than $1/2$ for uniform CTRs. That is, the CTR is either p or 0 (if the advertiser is not interested in this impression).

Progress made in the past decade on stochastic rewards is threefold. First, [MWZ14] gave the first algorithm that breaks the $1/2$ barrier for non-uniform but sufficiently small CTRs. On one hand, small CTRs are arguably the most relevant case in practice because most keywords' CTRs are less than 10%. On the other hand, 10% or even 1% is larger than the assumption made by [MWZ14] and its follow-up works. Hence, it remains an important open problem to design better online algorithms for less restrictive CTRs.

The second line of improvements comes from combining the online primal-dual framework and a more expressive linear program for the problem. This new analysis method gives a better understanding of classical algorithms BALANCE [HZ20] and RANKING [HJS+23] in the presence of stochastic rewards.

Researchers have also tried to gain new insight by considering a weaker clairvoyant benchmark. [GU23] showed that against the weaker benchmark, RANKING achieves the optimal $1 - 1/e$ competitive ratio for uniform CTRs. They also analyzed BALANCE for small CTRs, and obtained a ratio better than the aforementioned state-of-the-art against the offline optimum benchmark. The latter result was later improved by [HJS+23] to 0.611.

Last but not least, stochastic rewards are closely related to budget-oblivious algorithms for AdWords, i.e., algorithms which do not know an agent's budget until the moment it is depleted. By a reduction by [MWZ14], a competitive online algorithm for the latter model would yield the same competitive ratio in the former model (but not vice versa). Again, the greedy algorithm is a $1/2$-competitive budget-oblivious algorithm for this problem. The survey by [Meh13] listed finding a better budget-oblivious algorithm as an open problem. [Vaz23] suggested a variant of RANKING as a candidate algorithm. [LTX+23] showed that no variant of this algorithm is $(1 - 1/e)$-competitive. Finally, [Udw23] proved that the candidate is at least 0.508-competitive, and a variant of this algorithm is 0.522-competitive, both under the small-bids assumption, whereby the bids w_{ai}'s are small compared to the agent's budget B_a.

3 Beyond Online Bipartite Matching and Ad Allocation

The preceding online matching models, largely motivated by online advertising, crucially rely on the assumption that one side of the bipartite graph is fixed and known upfront. This prevents the theory of online matching being applied to other modern applications, including ride-hailing, ride-sharing, rental services, etc.

In this section, we discuss generalizations of classic online bipartite matching. The first two generalizations are motivated by ride-hailing and ride-sharing, that allow all vertices to arrive online and allow general (non-bipartite) graphs. The third is somewhat theoretical in nature, but is the most general problem. The last generalization is motivated by rental services and freelance labor markets, and so captures the reusability of resources.

	Fractional Relaxation	Original Problem
Fully Online	0.6 [TZ22]	0.569 [HTWZ20]
General Vertex Arrival	0.526 [WW15]	$1/2 + \Omega(1)$ [GKM+19]
Edge Arrival	$1/2$ [GKM+19]	$1/2$ [GKM+19]
Reusable Resources	$1 - 1/e$ [GIU21] $1 - 1/e$ [FNS21]	0.589 [DFN+23]

Table 2: State-of-the-art for online matching problems beyond bipartite matching & ad allocation.

3.1 Fully Online Model: Vertices with Arrivals and Deadlines

In online ride-hailing platforms (e.g., Uber, Lyft, DiDi), ride requests are submitted to the platform in an online fashion and are active in the system for a few minutes. The platform assigns each request to a currently available taxi (or self-employed driver). Requests and taxis can be modeled as vertices in a bipartite graph with edges between compatible ride requests and taxis. This is an online bipartite matching problem but does not fit into the classic model, since all vertices (both the requests and the taxis) arrive online. Similarly, ride-sharing platforms, which match ride requests (pairing up riders) are naturally modeled as an online matching problem on *general (non-bipartite) graphs*.

[HKT+20] introduced the fully online matching model to capture the above scenarios, though the same model was studied earlier by [BSZ06] in the context of liquidity in clearing markets. Let $G = (V, E)$ be the underlying graph, initially *completely unknown*. Each time step is either an arrival or a deadline of a vertex. Upon the arrival of a vertex, its incident edges to their previously-arrived neighbors are revealed. A vertex can be matched at any point until its *deadline*, with this time revealed on its arrival. Naturally, we assume the deadline of a vertex is after its arrival, and all edges incident to a vertex are revealed before its deadline. This model generalizes the classic one-sided online bipartite matching model, where all offline vertices arrive at the beginning and have deadlines at the end, and every online vertex has its deadline right after its arrival.

For the fully online matching problem, [HKT+20] proved that RANKING achieves a tight $\Omega \approx 0.567$ (the unique solution to $\Omega \cdot e^{\Omega} = 1$) competitive ratio for bipartite graphs, and a competitive ratio of 0.521 for general graphs. For the fractional variant of the problem, [HPT+19] established a tight $2 - \sqrt{2} \approx 0.585$ competitive ratio of BALANCE. Remarkably, RANKING and BALANCE are known to be optimal in the classic model, but the claimed tightness here only applies to the two algorithms themselves. Indeed, [HTWZ20] introduced the BALANCED-RANKING algorithm that achieves a competitive ratio of 0.569 for bipartite graphs, and the EAGER WATER-FILLING algorithm

that achieves a competitive ratio of 0.592 for the fractional variant. The later result was further improved by [TZ22] to 0.6. On the negative side, the state-of-the-art upper bound (i.e., hardness) is 0.613 [HKT+20, EKLS21, TZ22], separating the fully online model from the classic online bipartite matching model.

This setting is also known as the windowed online matching problem. [ABD+23] assumed a first-in-first-out structure on the active windows (i.e., arrivals and deadlines) of vertices, and achieved a $1/4$ competitive ratio for edge-weighted graphs through a reduction to the Display Ads problem by suffering an extra factor of 2. Combined with the state-of-the-art algorithm for the Display Ads problem by [BC21], their competitive ratio can be improved to 0.268.

3.2 General Vertex Arrival

Generalizing fully online matching is the online matching with general vertex arrivals problem, introduced by [WW15]. Again, the input is a graph $G = (V, E)$, initially unknown, with vertices arriving online. Upon the arrival of a vertex v, its incident edges to its previously-arrived neighbors are revealed. The algorithm either matches v to an unmatched neighbor immediately or leaves v unmatched, possibly matching it to a later-arriving neighbor u upon u's arrival. The inability to match vertices at any point before their departure (and lack of this information) makes this model more restrictive than the fully online model, and so algorithms for general vertex arrivals are also algorithms in the fully online model, with the same competitive ratio.

[WW15] presented a fractional 0.526-competitive algorithm for the fractional version of the problem. [GKM+19] designed a rounding of Wang and Wong's fractional algorithm and established a $1/2 + \Omega(1)$ competitive ratio for the integral matching problem. This result stands as the only non-trivial integral algorithm so far. On the negative side, [WW15, BST19, TWW22] established an upper bound of 0.583, separating the general vertex arrival model from the fully online model.

3.3 Edge Arrivals

Finally, we remark that the most general online matching setting is the edge arrival model. That is, edges of an underlying graph arrive in a sequence and the algorithm decides whether to select an edge immediately on its arrival. Here edges correspond to fleeting collaboration opportunities between agents. A competitive ratio of $1/2$ can be trivially achieved by a greedy algorithm that matches each edge on arrival if both its endpoints are free, and this is optimal for deterministic algorithms. Unfortunately, [GKM+19] proved that no online algorithm achieves a better than $1/2+1/2n$ competitive ratio, even for the fractional version of the problem. Positive results are known assuming structure, including low-degree graphs and trees [BST19], batching [LS17], random-order arrivals [GS17], or stochastic arrivals [GTW21] (see §4 for more on the latter models).

3.4 Reusable Resources

In sponsored search, the advertisers' budgets, viewed as resources, are non-reusable. In contrast, in such markets as cloud computing (e.g., AWS, Azure), short-term rentals (e.g., Airbnb), and freelancer labor (e.g., TaskRabbit), the allocated resources (be it compute, housing or labor) are reusable, and can be reallocated after being used.

The above motivates online bipartite matching with reusable resources, where after an offline vertex (a rental service) is matched, it becomes available again after d time steps, where d is a known parameter that corresponds to the usage duration of the vertices. The classic online bipartite matching problem is a special case of the reusable resources model when $d = \infty$.

This model was first introduced by [GGI+22] in a more general setting of online assortment optimization. [GIU21, FNS21] generalized the BALANCE algorithm and achieves an optimal $1-1/e$ competitive ratio for the fractional version of the problem.[2] For the integral version of the problem, [DFN+23] proposed the PERIODIC RERANKING algorithm (a variant of RANKING that reranks the offline vertices every d time steps), and show that it achieves a competitive ratio of 0.589, and an online correlated selection-based algorithm achieves a competitive ratio of 0.505. All these results extend to the vertex-weighted setting. We remark that the results of [DFN+23] heavily rely on the assumption that all vertices have identical usage durations d. The case of heterogeneous usage durations (i.e., each vertex v has an individual duration time d_v) remains open.

[FNS22] further studied online assortment of reusable resources in the stochastic setting. Reusable resources due to additional production have also been considered in infinite-horizon *stochastic* settings [AS20, CILB+20, KSSW22, PW24]. We discuss stochastic settings (without reusable resources) in more detail in the following section.

4 Stochastic Models: Secretaries, Prophets, and Philosophers

The preceding sections focused on adversarial models, where both input graph and arrival order are chosen by an adversary. This modeling choice, while robust, is quite pessimistic, and naturally results in worse guarantees than possibly achievable for real-world applications of interest. A natural way to obtain improved provable guarantees is to either consider random arrival orders (but adversarial input), or to posit a stochastic generative model, possibly learnt from historical data. Such models hearken back to classic results in optimal stopping theory concerning online Bayesian selection problems.

In the most basic setting, a buyer has a single item to sell, and impatient buyers arrive one after another and make take-it-or-leave-it bids for this single item. The buyer must select which bid to accept, immediately and irrevocably when the bid is made. Under adversarial models, a buyer cannot be competitive with the hindsight-optimal solution. In contrast, if the bids arrive in random order (referred to as the *secretary problem*), then a competitive ratio of $1/e$ is optimal [Dyn63]. Similarly, if the successive bids v_i are drawn independently from known distributions $\mathcal{D}_i$, then the optimal competitive ratio is $1/2$ [KS78], i.e., the buyer can guarantee an expected gain at least half of that obtained by a "prophet" who knows the realization of the randomness,

$$\mathbb{E}[\text{Gain}] \geq \frac{1}{2} \cdot \mathbb{E}[\max_i v_i].$$

Such guarantees contrasting with the offline optimal, or prophet, are referred to as *prophet inequalities*. One may also contrast with the (computationally-unbounded) optimal online algorithm for such problems, which for reasons elaborated below we refer to as *philosopher inequalities*.

These models have been generalized and extended significantly over the years. In this section, we focus on recent developments for generalizations of the above to bipartite matching markets, where the buyer wishes to sell multiple heterogeneous items, and each arriving buyer proposes a different bid for each item. Put otherwise, we focus on online bipartite matching models. We note that the buyer and seller terminology are not accidental, and these models have tight connections to questions in *mechanism design*, which we also discuss in this section.

[2] Equivalently, they assumed that resources (offline vertices) have large capacities.

	impossibility	algorithmic
Secretary Matching	$1/e$ [Dyn63]	$1/e$ [KRTV13]
Prophet Matching	$1/2$ [KS78]	$1/2$ [FGL15]
Philosopher Matching	0.99999 [PPSW21]	0.652 [NSW23]

Table 3: State-of-the-art for online bipartite weighted matching in stochastic settings.

4.1 Secretary Problems

For edge-weighted online bipartite matching with online vertices arriving in random order, [KP09] were the first to obtain a constant-competitive ratio, specifically a $1/8$-competitive algorithm. This was later improved to the optimal $1/e$ ratio by [KRTV13], generalizing the classic single-item result of [Dyn63], which we recall is the special case of a single offline vertex. For k heterogeneous offline vertices, [Kle05] showed $(1-1/\sqrt{k})$-competitive and truthful mechanism. [EFGT22a] study secretary matching in general graphs (with vertices arriving with edges to their previously-arrived neighbors, as in §3.1 and §3.2). They show that the optimal competitive ratio in this level of generality is $5/12$, notably strictly greater than achievable for bipartite graphs.[3]

The random-order model similarly allows for improved guarantees for the special cases of vertex-weighted and unweighted online bipartite matching. For unweighted matching, [GS17] show that for *edges* revealed in random order a better than $1/2$ competitive ratio is possible, notably beyond the worst-case optimal for adversarial arrivals [GKM+19]. Similarly, random-order vertex arrivals in bipartite graphs (with arrivals on only one side of the graph) allow one to surpass the worst-case optimal $1-1/e$: a generalization of RANKING achieves a competitive ratio of 0.662 [HTWZ19, JW21]. This generalizes results of [KMT11, MY11], who showed that for the unweighted problem RANKING (unchanged) achieves competitiveness beyond $1-1/e$, with the best known bound being 0.696 [MY11]. As noted by these last two works, these results for random-order arrivals imply the same competitive ratios for stochastic matching problems with unknown i.i.d. distributions over arrival types. This remains the best known result for unknown distributions. In the following sections, we discuss the types of guarantees achievable under *known* distributions.

4.2 Prophet Inequalities

The optimal competitive ratio of $1/2$ for the single-item problem due to [KS78] was also obtained several years later using a single-threshold (i.e., posted-price) algorithm by [SC84]. This results in truthful welfare-approximating mechanisms for single-item auctions. This connection between (pricing-based) prophet inequalities and mechanism design was later elaborated upon by researchers at the intersection of Economics and Computation [HKS07, CHMS10, KW19]. Interestingly, very recently [BHK+24] show that *any* guarantee achieved by an online Bayesian selection algorithm can be achieved by a (dynamic) posted-price policy, implying that studying the non-strategic setting results in truthful mechanisms which achieve the same approximation of the social welfare as the algorithm in strategic settings.

The connection between (combinatorial) prophet inequalities and mechanisms design continues to motivate a flurry of results on prophet inequalities for increasingly involved markets, with more and more sophisticated combinatorial constraints on the sets of buyers that may be serviced, or

[3] For some intuition as to why this is not a contradiction, note that in the star example (i.e., the single-item problem), the center of the star arrives after the highest-bidding neighbor with probability $1/2 > 1/e$, and so greedily matching the center when it arrives is $1/2$-competitive, and the lower bound of [Dyn63] for bipartite graphs does not apply.

items sold. See the excellent surveys [HK92, CFH+18] and [Har12, Luc17]for more on prophet inequalities and their connection to mechanism design, respectively. In what follows, we focus on prophet inequalities subject to matching constraints.

For unweighted online bipartite matching, [FMMM09] were the first to show that stochastic inputs allow for competitive ratio beyond the worst-case optimal $1 - 1/e \approx 0.632$. Specifically, they show that if online vertices' neighborhoods are drawn i.i.d. from a single known distribution, then a competitive ratio of 0.67 is achievable. There has been a long line of work studying this question, most recently [JL13, BGMS21, HS21, HSY22, TWW22], with the current best competitive ratios being 0.7299 and 0.716 assuming integral and arbitrary arrival rates [BGMS21, HSY22].[4] For edge-weighted matching, a number of results were obtained under integral arrival rates [HMZ11, BGMS21], with the best ratio standing at 0.704, while for arbitrary arrivals the ratio of $1 - 1/e$ was only recently beaten [Yan24, QFZW23]. In contrast, by a work of [MGS12], no competitive ratios greater than $1 - 1/e^2 \approx 0.864$ and 0.823 are possible in the same settings.

For unweighted and vertex-weighted bipartite matching under (much more general) time-varying independent distributions, [TWW22] recently provided the first algorithm surpassing the competitive ratio of $1 - 1/e$, presenting a 0.666-competitive algorithm. For edge-weighted matching a competitive ratio of $1/2$ is best possible, as this generalizes the single-item problem of [KS78]. This ratio is known to be achievable via numerous approaches [FGL15, DFKL20, EFGT22b]. This ratio of $1/2$ is even achievable under vertex arrivals in general graphs [EFGT22b], or correlated arrivals [AM23]. In contrast, the problem is strictly harder under edge arrivals, where the best known competitive ratio is in the range $[0.344, 0.4]$ [MMG23] and $[0.349, 3/7]$ for bipartite graphs [MMG23, CCF+23]. For unweighted matching a competitive ratio of 0.502 is possible [GTW21].

4.3 Philosopher Inequalities

While a competitive ratio of $1/2$ is worst-case optimal for online Bayesian selection subject to bipartite matching constraints, this is still a pessimistic worst-case guarantee, as the lower bounds focus on worst-case distributions. The optimal algorithms for distributions of interest may allow for better competitive ratios. This optimal algorithm, which is the solution of a Markov Decision Problem (MDP), is computable in polynomial space via standard techniques. As shown by [PPSW21], this is the right characterization, and even *approximating* the optimal policy beyond some 0.999 ratio is PSPACE-complete (i.e., is as hard as the hardest problems requiring polynomial space). Hence, under standard complexity-theoretic assumptions, this optimal policy is not computable in polynomial time. Put otherwise, it is likely computable only by a character with sufficient time to "think" (i.e., compute), therefore naturally referred to as a "philosopher". This motivates the study of polynomial-time approximation of the optimal online algorithm, which, in analogy with prophet inequalities (approximation of the optimal offline algorithm), we term *philosopher inequalities*.

[ANSS19] were the first to consider the approximation of the optimal online algorithm for online Bayesian selection. They considered bounded-depth and production-constrained laminar matroids, for which they provided $(1+\epsilon)$-approximate philosopher inequalities for any constant $\epsilon > 0$. [DGR+23] obtained the same bounds for random-order (secretary) philosopher inequalities for a single item. For online bipartite matching, which by [PPSW21] such an approximation would result in surprising developments in complexity theory. On the positive side, a successive line of work [PPSW21, SW21, BDL22, NSW23] showed that (increasingly) better than $1/2$-approximate philosopher inequalities are possible. The current best bound stands at 0.652 (notably, above the natural bound of $1 - 1/e$ for online matching algorithms). We note that the recent work of [BHK+24] also

[4] The arrival rate is the expected number of arrivals of a particular online type.